Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 1 of 7 · Question 2 part d · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

2(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • (Touch)screen (1) • Speaker (1) • Vibration (motor) (1) • Flashlight/LED light/torch (1) • NFC chip (1)	**Accept:** • display for screen **Do not accept:** • monitor • printer • headset • other peripheral devices	**2**
Question	Answer	Additional Guidance	Mark
2(b)(i)	**The only correct answer is D** *A is not correct because cache is located closer to the CPU* *B is not correct because random-access memory is a physical card on the motherboard* *C is not correct because read-only memory cannot be written to*		**1**
Question	Answer	Additional Guidance	Mark
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Often held/stored remotely/stored on the internet (1) • Held/managed by a third party (1) • Requires internet connectivity to access (1)		**2**

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 1 of 7 · Question 2 part d · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for any of the following up to a maximum of **three** marks: • Clock speed/number of cycles per second (1) • Number of processor cores (1) • Size of cache (memory) (1) • Type of cache (1) • Width of data bus/word length (1) • Width of address bus (1) • Whether the CPU is overclocked or not (1) • Pipelining architecture (1) • Thermal throttling / cooling system (reducing clock speed) (1)	**Accept:** • units of clock speed e.g. GHz for MP1 • width of bus for either MP5 or MP6 but not both **Do not accept:** cache without clarification of size or type	3
Question	Answer	Additional Guidance	Mark
2(c)(ii)	Award up to **two** marks for a linked description such as: • The CPU places an address on to the address bus (1) to specify the location in main memory (1) • (The address bus) contains the location in RAM (1) to read from/write to (1) • The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1) • The address bus carries the memory location (1) of the current/next instruction (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for any of the following up to a maximum of **two** marks: • Sub-atomic particles (1) • Qubit(s) (1) • Entanglement (1) behaviour of a pair of qubits can influence each other • Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously	**Do not accept:** 'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question	2
		Total for question 2	**12**

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 1 of 7 · Question 2 part d · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question	Answer	Additional Guidance	Mark

6(a)(i)	Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1)		2

Question Number	Answer	Additional Guidance	Mark
6(a)(ii)	Any **two** from: • Printer (1) • Bank card (magnetic strip) reader/scanner (1) • Bank card contactless/NFC reader/scanner (1) • Cash input and giving change device (1)	Allow any hardware that relates to paying by cash for MP4	2

Question Number	Answer	Additional Guidance	Mark
6(a)(iii)	To prevent unauthorised people from reading / understanding the bank card information (1)	Do not award 'to keep it from being hacked / accessed'.	1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 2 of 7 · Question 6 part c · 6 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for each of (maximum of 4): 1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1) 2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1) 3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1) 4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1) 5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1)	For each mark point: • the general feature is followed by exemplification • needs to be a comparison, which may be implicit	4

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 2 of 7 · Question 6 part c · 6 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

Question Number	Answer	Mark
6(c)	Indicative content: Characteristics – Any of the following: • Implemented in software. • Could involve the use of neural networks / clustering / modelling • Algorithms are trained by exposure to data that has been validated / checked • Algorithms learn by identifying commonalities between the validated / checked data • Algorithms make predictions for new data based on prior learning • Algorithms may incorporate recursive self-improvement Uses – Any of the following: • Game playing (opponents) • Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions) • Image / Audio processing (recognising objects/patterns, face recognition) • Logistics (scheduling, order fulfilment) • Control systems (cars, manufacturing, weapons, navigation • Diagnostic systems (medical, mechanical, electrical) • Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants) • Robotics (dangerous situations, help aged or disabled) Ethical issues – Any of the following: • Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes • Is a computer to be trusted to make decisions (life-death)? • If a computer discovers something that humans can't prove, should it be accepted as truth? • How do we know that the answer produced is correct? • Will people be comfortable interacting with machines that are considered intelligent? • Some jobs may be at risk, especially those requiring human skills such as empathy or creativity.	6

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 2 of 7 · Question 6 part c · 6 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 2 of 7 · Question 6 part c · 6 marks

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Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 3 of 7 · Question 6 part a.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

6(a)(i)	Award **two** marks for a linked explanation such as: • Uses DNA/biomolecular components (1) rather that standard hardware/silicon chips (1) • DNA uses four-character genetic alphabet (1) rather than binary digits (1) • DNA uses chemical reactions (1) rather than electrical states/properties (1) • DNA computing uses massive parallel processing/strands (1) to solve problems that otherwise would take impossible amounts of time (1)		2
6(a)(ii)	Award **one** mark from: • The ability of a system to be in multiple states at the same time (1) • Multiple probabilities at the same time (1) • One state combines all possibilities (1) • The state is not known until it is measured (1)		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 3 of 7 · Question 6 part a.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **two** marks from: • Some people may not want to use technology (1) • Some people may not be able to afford to use technology (1) • Some people may not have access to the technology/infrastructure (1) • Some people may not be allowed access to technology (1) • Some people may not be able to use it (disabilities etc.) (1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)	Indicative content Impacts • Pollution (water, air, noise) resulting from the manufacturing process • High energy and water volumes needed for the manufacturing process • Mineral mining contaminates ground water • Mineral (copper, gold, silver, lithium) resource depletion • High energy use to keep machines cool with air conditioning or fans • Contain toxic components which means computing devices are hazardous waste • Batteries (laptop, lithium cells) disposal is hazardous and needs specialised disposal facilities • Computing devices sent to landfills contaminate ground and water resources Actions • Check national and local legislation • Some countries/states/regions require sellers to recycle own waste • Research recycling facilities to see if e-waste is acceptable • Find if local/national/regional government agencies charge for collection and disposal of e-waste • Find sellers offering exchange old for new + cost purchase options • Find charities (local, regional) which take donations of old devices • Turn off machines when not in use to save energy		6

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 3 of 7 · Question 6 part a.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content
Level 1	1-2	Basic independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 3 of 7 · Question 6 part a.i · 2 marks

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Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 4 of 7 · Question 6 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

6(a)(i)	Award **two** marks for a linked explanation such as: • Uses DNA/biomolecular components (1) rather that standard hardware/silicon chips (1) • DNA uses four-character genetic alphabet (1) rather than binary digits (1) • DNA uses chemical reactions (1) rather than electrical states/properties (1) • DNA computing uses massive parallel processing/strands (1) to solve problems that otherwise would take impossible amounts of time (1)		2
6(a)(ii)	Award **one** mark from: • The ability of a system to be in multiple states at the same time (1) • Multiple probabilities at the same time (1) • One state combines all possibilities (1) • The state is not known until it is measured (1)		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 4 of 7 · Question 6 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **two** marks from: • Some people may not want to use technology (1) • Some people may not be able to afford to use technology (1) • Some people may not have access to the technology/infrastructure (1) • Some people may not be allowed access to technology (1) • Some people may not be able to use it (disabilities etc.) (1)		2

Question Number	Answer	Additional Guidance	Mark
6(c)	Indicative content Impacts • Pollution (water, air, noise) resulting from the manufacturing process • High energy and water volumes needed for the manufacturing process • Mineral mining contaminates ground water • Mineral (copper, gold, silver, lithium) resource depletion • High energy use to keep machines cool with air conditioning or fans • Contain toxic components which means computing devices are hazardous waste • Batteries (laptop, lithium cells) disposal is hazardous and needs specialised disposal facilities • Computing devices sent to landfills contaminate ground and water resources Actions • Check national and local legislation • Some countries/states/regions require sellers to recycle own waste • Research recycling facilities to see if e-waste is acceptable • Find if local/national/regional government agencies charge for collection and disposal of e-waste • Find sellers offering exchange old for new + cost purchase options • Find charities (local, regional) which take donations of old devices • Turn off machines when not in use to save energy		6

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 4 of 7 · Question 6 part a.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content
Level 1	1-2	Basic independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 4 of 7 · Question 6 part a.ii · 1 mark

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Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 5 of 7 · Question 5 part a.i · 2 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part a.i in this mark scheme extract

5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 5 of 7 · Question 5 part a.i · 2 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: • Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: • Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** *A is not correct because this is phishing* *B is not correct because this is shoulder surfing* *D is not correct because this is pharming*		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 5 of 7 · Question 5 part a.i · 2 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part a.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award up to two marks for a linked explanation such as:

• Robust/less likely to break it if dropped / quieter than a mechanical hard drive (1) because it doesn't have any moving/mechanical parts (1)
• Smaller/thinner/lighter than a device with HDD (1) so easier to carry/more portable / laptops have less room in them for drives/componants(1)
• SSD uses less power (1) so battery will last longer (1)
• SSD is faster (1) so can access data/records more quickly (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(ii)</td><td>Any two from:

• Flash memory chips are used (1)
• Chips have (floating gate) transistors/electron pools/charge traps (1)
• Gates/electron pools/charge traps hold an electrical charge (1)
• Charge remains even when no power (1)
• No charge/empty pool/trap represents data/1 (1)
• A charge/full pool/trap represents no data/0 (1)
• Data is stored in blocks and pages (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(iii)</td><td>Award one mark for each of:

<table><tr><th>Description</th><th>RAM</th><th>ROM</th></tr><tr><td>Stores the boot up sequence</td><td></td><td>▯</td></tr><tr><td>The contents are lost when the laptop is shut down</td><td>▯</td><td></td></tr></table></td><td></td><td>2</td></tr>
<tr><td colspan="3">Total for question 5</td><td>12</td></tr>
</table>

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 5 of 7 · Question 5 part a.i · 2 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 6 of 7 · Question 4 part d.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part d.i in this mark scheme extract

Number			
4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 6 of 7 · Question 4 part d.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part d.i in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 6 of 7 · Question 4 part d.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part d.i in this mark scheme extract

4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 6 of 7 · Question 4 part d.i · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part d.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 6 of 7 · Question 4 part d.i · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part d.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 7 of 7 · Question 4 part d.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 5 · find part d.ii in this mark scheme extract

Number			
4(a)	Award up to **two** marks for a linked explanation such as: • 'What-if' questions can be asked by changing variables (1) to make predictions / see trends (1) • The simulation can be rerun with different values (1) to see how the outcomes will change (1) • The simulation can be run faster than real life (1) so multiple generations of animal results can be seen / long-term trends seen (1) • The simulation can be safer (1) because of the danger of animal interaction (1) • The simulation would be more convenient / saves time (1) because observers would have to be present outside for a long time (1)	Looking for a relevant statement and a linked extension/expansion. Do not award two independent statements that are not linked. Points could be reversed such as: you can make predictions because variables can be changed to see how the outcome will differ.	2
4(b)	**The only correct answer is C** ***A** is not correct because it is not a linear scenario* ***B** is not correct because parallel models do not have to communicate and coordinate their actions* ***D** is not correct because there is not an explicit input with an expected output*		1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 7 of 7 · Question 4 part d.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 5 · find part d.ii in this mark scheme extract

4(c)	Award up to **two** marks for a linked description such as: • The research facility could use Two/Multi Factor Authentication (2FA/MFA) (1) sending a secondary piece of data to the employee which they then enter (e.g. passcode / authenticator app / email) (1) • Using a username and password (1) that is unique to each individual employee / that matches one stored in a database (1) • The research facility could implement a security question (1) that only the employee would know the answer to (1)	Do not allow any biometric measure e.g. fingerprint - biometrics is given in the question. Do not accept password on its own - this is insufficient to allow the user's identity to be authenticated	2
4(d)(i)	Award **one** mark for any of the following: • The manipulation of matter with a size from 1 to 100 nanometres / scale of 10^{-9} / at atomic/molecular level (1) • The inclusion of nanoparticles into different materials (1) • Facilitates miniaturisation at atomic/molecular **scale/level** (1)	Small/miniature/microscopic is not enough. Responses must relate to the scale at the nano/molecular/atomic level.	1
4(d)(ii)	Award **one** mark for any of the following: • Stronger materials e.g. scratch resistant coatings / self-cleaning glass (1) • Miniaturisation of electronics e.g. transistors (1) • Nanomedicines to deliver drugs (1)	Allow examples that reasonably map to the categories given.	1

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 7 of 7 · Question 4 part d.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 5 · find part d.ii in this mark scheme extract

4(e)	Award **one** mark for each of the following up to a maximum of **three** marks: • It uses a total of 128 bits (1) • There is a total of 32 hexadecimal character (1) • Each hexadecimal character requires 4 bits (1) • There are 8 blocks of 4 character / 8 x 4 blocks (1) • Each block (of 4 character) requires 16 bits (1) • The address uses 4 (bits per character) x 32 (characters) / 4 x 32 (1) • Each of the 8 blocks uses 16 bits / 8 x 16 (1)		3
4(f)	Indicative content • Rights and responsibilities • Data Protection regarding personal data • Surveillance • Consent of the individual to having specified data held • Accountability of those running / using such systems • Ownership • Data mining & analysis of work patterns and potential use/misuse • Ethics versus the legality of implementing such systems • Data misuse and protection against hacking and malware • Requirement to keep data secure • Physical access to areas of the site • Monitoring employee movement to restricted areas / resources • Algorithmic bias in facial recognition		6
		Total for question 4	**16**

Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 7 of 7 · Question 4 part d.ii · 1 mark

2024 JUN · Paper 01 · extract 4 of 5 · find part d.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
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Type MS · Emerging tech: quantum, DNA computing, AI and nanotechnology

Answer 7 of 7 · Question 4 part d.ii · 1 mark

2024 JUN · Paper 01 · extract 5 of 5 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark